Auditing Surface-Order Approaches to GRH: Detailed Proofs and Research Interfaces

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Research draft. GRH is not proved here.

Abstract

We audit proposed surface-order and Hilbert–Pólya arguments for GRH. We retain a conditional spectral-approximation criterion and prove three obstructions: integral hyperbolic matrices have no integer prime-power norms, integral cyclic induction fails already for the Klein four group, and absolute prime tails diverge throughout the proposed critical-strip range. We also exhibit determinant information lost by symmetrization and a direct failure of the supplied diagonal diagnostic. We supply convergent arithmetic replacements using Dirichlet partial sums and pole-subtracted zeta sums, with explicit errors. These results identify what a complete arithmetic spectral construction must still establish. They do not establish GRH. A Lean 4 companion verifies exact arithmetic obstructions, complex Euler-factor nonvanishing, and a local boundary-convergence lemma. The full arithmetic spectral bridge remains unresolved.

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1 Scope and conventions

For a primitive Dirichlet character χ of conductor q, write

$$L(s,\chi)=\sum_{n\geq 1}\frac{\chi(n)}{n^s},\qquad \Lambda(s,\chi)=\left(\frac{q}{\pi}\right)^{(s+a)/2}\Gamma\left(\frac{s+a}{2}\right)L(s,\chi),\quad a=\frac{1-\chi(-1)}{2}.$$

The series and Euler product initially apply for $\operatorname{Re} s>1$. Our target is Dirichlet GRH: every zero in $0<\operatorname{Re} s<1$ has real part $1/2$. For $q=1$, use the entire function $\xi(s)=\frac{1}{2}s(s-1)\Lambda(s,1)$ when an entire target is required. Finite-order Hecke characters and general Artin representations require separate constructions; a theorem for one family is not silently a theorem for another.

The supplied surface-order manuscripts [1, 2] propose arithmetic transfer operators and self-adjoint approximants. We isolate valid closure logic and prove obstructions to particular claimed constructions. These obstructions refute intermediate statements, not GRH. The separate `math` repository [3] includes claimed zero-free regions strictly weaker than GRH. Such a region cannot replace the missing critical-line argument.

2 A corrected conditional closure theorem

Theorem 2.1 (Spectral approximation criterion)**.** *Let $U\subset\mathbb{C}$ be connected and open, let f be holomorphic on U and not identically zero, and let*

$$F_N(s)=e^{a_N+b_N s}\det(sI-(\tfrac{1}{2}I+iH_N)),\qquad H_N=H_N^*\in M_{d_N}(\mathbb{C}).$$

If $F_N\to f$ locally uniformly on U, then every zero of f in U has real part $1/2$.

Proof. The eigenvalues $\lambda_{j,N}$ of H_N are real. Thus the zeros of F_N are $1/2+i\lambda_{j,N}$, with algebraic multiplicities; the exponential factor never vanishes. If $f(z)=0$ off that line, choose a closed disk $D\subset U$ about z disjoint from the line, with f nonzero on its boundary. Such a disk exists because the zeros of a nonzero holomorphic function on a connected domain are isolated. Let $m=\min_{\partial D}|f|>0$. Local uniform convergence gives $|F_N-f|<m$ on ∂D for sufficiently large N. Rouché's theorem forces F_N and f to have the same positive number of zeros inside D, contradicting the zero location of F_N. □

Remark 2.2. This is a conditional criterion, not a construction. Self-adjointness alone establishes only the zero location of the approximants. The difficult arithmetic statement is convergence to the particular completed L-function. Pointwise convergence, weak operator convergence, and agreement of finitely many zeros do not supply the displayed hypothesis.

Proposition 2.3 (A usable continuation interface)**.** *Suppose U is a connected domain, the entire functions F_N are locally uniformly bounded on U, and they converge pointwise on a nonempty open subset $V\subset U$ to a holomorphic function f defined on U. Then $F_N\to f$ locally uniformly on U.*

Proof. Montel's theorem makes the family normal. Every subsequence has a further subsequence converging locally uniformly to a holomorphic function g. On V, pointwise convergence gives $g=f$, so the identity theorem gives $g=f$ on U. If the original sequence failed to converge locally uniformly, a compact set, positive error, and violating subsequence would contradict this subsequence property. □

A proposed analytic continuation proof must actually provide the uniform bounds and a genuine overlap V. The disjoint domains $\operatorname{Re} s>1$ and $0<\operatorname{Re} s<1$ are not an overlap.

3 Arithmetic obstructions to the claimed completion

Theorem 3.1 (No integral prime-power geodesic norms)**.** *Let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ be hyperbolic. Its expanding eigenvalue in absolute value is $u > 1$, and its hyperbolic norm is $N(\gamma) = u^2$. Then $N(\gamma)$ is not an integer greater than one. In particular no closed geodesic arising from such a matrix has length $k \log p$ for a prime p and integer $k \geq 1$.*

Proof. Put $t = |\operatorname{tr} \gamma| \in \mathbb{Z}$, so $t = u + u^{-1}$. If $u^2 = n \in \mathbb{Z}$, $n > 1$, squaring gives

$$t^2 = n + 2 + \frac{1}{n}.$$

The left-hand side and $n + 2$ are integers, so $1/n$ is an integer, contradicting $0 < 1/n < 1$. Since length is $\log N(\gamma)$, length $k \log p$ would imply $N(\gamma) = p^k$, already excluded. □

The statement applies to every subgroup of $\mathrm{SL}_2(\mathbb{Z})$, hence to congruence surfaces. The prime geodesic theorem is an asymptotic counting analogy; it cannot provide the termwise identification claimed in the completion section of [1]. Hecke correspondences of determinant p must not be identified with hyperbolic deck transformations of determinant one.

Theorem 3.2 (Failure of integral cyclic induction)**.** *The trivial character of $G = C_2 \times C_2$ is not an integer linear combination of characters induced from one-dimensional characters of cyclic subgroups of G.*

Proof. Every cyclic subgroup has order one or two. The degree of any such induced character is its index, namely four or two. Evaluation at the identity sends every integer linear combination to an even integer, whereas the trivial character has degree one. □

This distinguishes rational Artin induction from integral Brauer induction. It invalidates the stated cyclic integral reduction, while leaving appropriate forms of these classical induction theorems available for a revised argument.

Theorem 3.3 (Prime-tail obstruction)**.** *For every real $\sigma \leq 1$, the positive series $\sum_p p^{-\sigma}$ diverges. Thus $\sum_{p>N} p^{-\sigma}$ is not a vanishing absolutely convergent tail in the critical strip.*

Proof. It suffices to show $\sum_p 1/p = \infty$, since $p^{-\sigma} \geq p^{-1}$. Suppose the reciprocal-prime series converged. For $0 \leq x \leq 1/2$, $-\log(1 - x) \leq 2x$, for example by integrating $(1 - t)^{-1} \leq 2$ from zero to x. Hence the finite Euler products $\prod_{p \leq X}(1 - 1/p)^{-1}$ would be bounded uniformly in X. Expanding each as a finite product of absolutely convergent geometric series gives the sum of $1/n$ over all positive integers with prime factors at most X. In particular it dominates $\sum_{n \leq \lfloor X \rfloor} 1/n$, which is unbounded. Contradiction. □

Any repair below $\operatorname{Re} s = 1$ must introduce proved cancellation or a regularization, with quantitative control; absolute prime-tail bounds cannot suffice.

4 Determinants and lost arithmetic information

For $T = \operatorname{diag}(\chi(p)p^{-s})_{p \leq N}$ one has exactly

$$\det(I - T) = \prod_{p \leq N} (1 - \chi(p)p^{-s}) = L_N(s, \chi)^{-1},$$

not L_N. On an absolute-convergence region,

$$\log\det(I-T) = -\sum_{k\geq 1}\frac{\operatorname{tr}(T^k)}{k}.$$

The reciprocal orientation matters before any analytic continuation. Moreover Hermitian-part extraction does not preserve this determinant:

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \frac{T+T^*}{2} = 0, \quad \det(I-T) = 2 \neq 1 = \det(I).$$

An identity for the transfer determinant therefore does not automatically identify the centered characteristic polynomial of a symmetrized operator.

Proposition 4.1 (A direct obstruction to the supplied diagonal diagnostic)**.** *For the $q = 1$ model evaluated at $s = 1/2$, the algorithm that sets $\lambda_p = -\arg(p^{-1/2})$ gives $H_N = 0$ and $F_N(s) = (s-1/2)^{d_N}$. These functions cannot converge locally uniformly to $\Lambda(s,1)$ or $\xi(s)$ on a neighborhood of $1/2$.*

Proof. Every $p^{-1/2}$ is positive, so its argument is zero and $F_N(1/2) = 0$. On the other hand the alternating Dirichlet series $\eta(1/2) = \sum_{n\geq 1}(-1)^{n-1}n^{-1/2}$ is positive: grouping into adjacent pairs gives positive terms and its sum is at least $1 - 1/\sqrt{2} > 0$. The continuation identity $\eta(s) = (1-2^{1-s})\zeta(s)$ gives $\zeta(1/2) < 0$. The gamma factor at $1/4$ is positive and finite, so $\Lambda(1/2,1) \neq 0$, and its pole-removal factor at $1/2$ is nonzero. Even pointwise convergence at that point is impossible. □

Proposition 4.2 (Holomorphic Hermitian rigidity)**.** *Let $U \subset \mathbb{C}$ be connected and open, with a fixed inner product on $\mathbb{C}^d$. If $H : U \to M_d(\mathbb{C})$ has holomorphic entries and $H(s)^* = H(s)$ for every complex $s \in U$, then H is constant.*

Proof. In fixed orthonormal coordinates, for each vector v, the holomorphic scalar $v^*H(s)v$ is real-valued, so it is constant by the open mapping theorem. Polarization recovers all matrix entries from these quadratic forms, hence every entry is constant. Equivalently, $H_{ij} + H_{ji}$ and $i(H_{ij} - H_{ji})$ are real-valued holomorphic functions whose constancy determines H_{ij}. □

Thus a nonconstant arithmetic transfer $T(s)$ cannot retain entrywise holomorphy after pointwise Hermitian-part extraction: a holomorphic $(T+T^*)/2$ would be constant, and the remaining holomorphic skew-Hermitian part would be constant as well. This excludes that particular shortcut, not fixed self-adjoint operators. The spectral criterion uses H_N independent of the spectral variable s. Requiring self-adjointness only on a real parameter line is also consistent with a nonconstant holomorphic extension, but needs its own zero-location theorem.

Proposition 4.3 (A positive generalized spectral pencil)**.** *Let $A = A^*$ and $B = B^* > 0$ be fixed $d \times d$ matrices. Every zero of $\det(A - zB)$ is real. Consequently every zero of $\det(A + i(s-1/2)B)$ lies on $\operatorname{Re} s = 1/2$.*

Proof. Singularity supplies $v \neq 0$ with $Av = zBv$. Hence $z = (v^*Av)/(v^*Bv)$, with real numerator and strictly positive real denominator. Thus z is real. Substitute $z = -i(s-1/2)$. □

This supplies a valid finite spectral mechanism, but not its arithmetic construction. To apply it one must construct A_N, B_N, certify $B_N > 0$, and prove the required completed-L determinant approximation independently. Hermitian coefficients alone are insufficient: for

$$A = \begin{pmatrix} 0 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \det(A - zB) = -(z^2 + \varepsilon^2),$$

$0 < \varepsilon < 1/2$ gives transported off-line zeros $s = 1/2 \pm \varepsilon$ inside the critical strip.

5 A necessary change of approximation class

Theorem 5.1 (Zero-free Euler truncations cannot approximate a zero)**.** *Let U be a connected open subset of $\{\operatorname{Re} s > 0\}$, let g_N be holomorphic and nowhere zero on U, and let P_N be arbitrary finite sets of rational primes. Put*

$$D_N(s) = \prod_{p \in P_N} (1 - \chi(p) e^{-s \log p}).$$

Each of $g_N D_N$ and g_N / D_N is holomorphic and nowhere zero on U. If either sequence converges locally uniformly to a holomorphic f which is not identically zero, then f has no zero in U.

Proof. For $\operatorname{Re} s > 0$, $|\chi(p) e^{-s \log p}| \le p^{-\operatorname{Re} s} < 1$, so every local factor is nonzero. Hence the finite product, its reciprocal, and their normalized versions have the asserted property. If f had a zero, choose a closed disk inside U with a positive number of zeros of f in its interior and none on its boundary. Local uniform convergence and Rouché's theorem would force the zero-free approximants to have that positive number of zeros, a contradiction. □

More quantitatively, on any such disk D, every zero-free holomorphic approximant G satisfies

$$\max_{\partial D} |G - f| \;\ge\; \min_{\partial D} |f| > 0.$$

Otherwise Rouché yields the same contradiction. Thus improved estimates cannot repair this approximation class near a nontrivial zero. The construction must change so that finite approximants can have zeros in the strip; a nonvanishing gamma or exponential normalization cannot do this.

6 A convergent arithmetic replacement

For a nonprincipal character modulo q, put $A(x) = \sum_{n \le x} \chi(n)$ and $B = \sup_x |A(x)| \le q$. Complete periods sum to zero, so B is finite. Partial summation gives, for $\sigma = \operatorname{Re} s > 0$,

$$L(s, \chi) - \sum_{n \le N} \chi(n) n^{-s} = -A(N) N^{-s} + s \int_N^\infty A(x) x^{-s-1} \, dx,$$

and therefore

$$\left| L(s, \chi) - \sum_{n \le N} \chi(n) n^{-s} \right| \le B \left(1 + \frac{|s|}{\sigma} \right) N^{-\sigma}.$$

The endpoint tends to zero and the integral is absolutely convergent. The limit agrees with the Euler definition on $\operatorname{Re} s > 1$, hence with its analytic continuation. Multiplication by the correct gamma/conductor factor gives locally uniform approximation on $\operatorname{Re} s > 0$ for either parity.

For zeta, reproduce the pole before removing it. Set

$$Z_N(s) = \sum_{n \le N} n^{-s} + \frac{N^{1-s}}{s-1}.$$

Euler summation gives, for $\sigma > 0$ and $s \neq 1$,

$$\zeta(s) - Z_N(s) = -s \int_N^\infty \{x\} x^{-s-1}\, dx, \qquad |\zeta(s) - Z_N(s)| \le \frac{|s|}{\sigma} N^{-\sigma}.$$

The integral continues holomorphically to $\sigma > 0$. Consequently

$$\xi_N(s) = \frac{s}{2}\pi^{-s/2}\Gamma(s/2)\left((s-1)\sum_{n \le N} n^{-s} + N^{1-s}\right)$$

is holomorphic on that half-plane, including $s = 1$, and converges locally uniformly to ξ; in fact $\xi_N(1) = 1/2$ for every N. These constructions use known arithmetic coefficients, not unknown zeros. They avoid Theorem 5.1 by allowing additive cancellation and unrestricted finite-level zeros. They establish convergence, but provide no self-adjoint realization or critical-line theorem for those zeros.

7 A certified test of the arithmetic pencil candidate

Let χ be the primitive nonprincipal character modulo 3. Its fifth Dirichlet partial sum is the entire function

$$f(s) = S_5(s, \chi) = 1 - 2^{-s} + 4^{-s} - 5^{-s}.$$

Theorem 7.1 (Certified zero of a finite partial sum)**.** *Put*

$$c = (337261 + 11027504i)/10^6, \qquad r = 10^{-4}.$$

The disk $|s - c| < r$ contains exactly one zero of f, counted with multiplicity, and lies inside $0 < \operatorname{Re} s < 1/2$.

Proof. The exact rational procedure detailed in the extended edition establishes

$$|f(c)| < 10^{-6}, \qquad |f'(c)| > 1, \qquad \sup_{|s-c| \le r} |f''(s)| \le 9.$$

It uses 60 terms of the logarithm's positive atanh series and terms 0 through 120 of the exponential Taylor series, with explicit remainders; no floating-point value is used in these inequalities. On the boundary, Taylor's formula gives

$$|f(s) - f'(c)(s-c)| < 10^{-6} + \tfrac{9}{2}r^2 < r < |f'(c)(s-c)|.$$

Rouché gives exactly one zero, which is therefore simple. The real part throughout the closed disk belongs to $[337161, 337361]/10^6 \subset (0, 1/2)$. □

The completed fifth partial sum has the same zero because its conductor/gamma factor is holomorphic and nonzero on this disk. It cannot equal a nowhere-zero holomorphic factor times the determinant of a fixed positive Hermitian pencil there. This rejects that exact finite-level identification. It does not reject convergence of the partial sums, modified approximants, later truncations, subsequences, or GRH itself: f is a finite partial sum, not the full L-function.

8 Consequences for a revised research article

Theorems 3.1, 3.2, and 3.3 require replacements for the associated proof steps, not additional exposition. A credible next theorem would construct operators independently of unknown zeros and establish their exact arithmetic determinant, including ramified, archimedean, and conductor terms. It must then prove locally uniform convergence or the local boundedness in Proposition 2.3. Only then does Theorem 2.1 prove the desired zero location.

Weil/Li positivity, adelic trace methods, and canonical-product approximations are potential alternative interfaces. They move the research obligation to a specific positivity, trace, or approximation theorem; they do not discharge it by analogy. Lagarias [4] and Connes [5] provide relevant primary-source starting points. No novelty or complete proof is claimed for these classical conditional principles.

A Worked geodesic calculations and possible replacements

For trace $t = 3$, the expanding eigenvalue is $(3 + \sqrt{5})/2$ and

$$N = \frac{7 + 3\sqrt{5}}{2}, \qquad \ell = 2 \log \frac{3 + \sqrt{5}}{2}.$$

This is a legitimate hyperbolic length, but not $\log p$. For $p = 2$, the proposed length $\log 2$ would require $t = 3/\sqrt{2}$, equivalently $2t^2 = 9$, which is impossible even modulo two. The Lean companion verifies this last exact integer obstruction. For arbitrary p^k, the integer argument in Theorem 3.1 is stronger than a numerical search over traces.

A Hecke operator attached to determinant-p double cosets is a different object from a determinant-one deck transformation. A replacement must specify the ambient space, coefficient system, trace class or regularized trace, and the precise arithmetic distribution. Reusing the same symbol for these objects does not construct an identification. An adelic framework may encode prime norms intrinsically; its trace and positivity obligations must then be proved in that framework.

B Character examples and induction bookkeeping

The nonprincipal character modulo three has values $\chi(1) = 1$, $\chi(2) = -1$, $\chi(0) = 0$ and is odd. For scalar weight-zero automorphy under $\Gamma_0(3)$,

$$f(\gamma z) = \chi(d) f(z),$$

the element $-I$ fixes z and has $\chi(-1) = -1$. Hence every such function would satisfy $f(z) = -f(z)$ and vanish. The proposed scalar bundle on the projective quotient is therefore unavailable as written. One can work on a different group or choose a compatible nonzero weight; either change requires rebuilding the operator and trace identity.

Modulo five, the quadratic character has $\chi(2) = -1$. Consequently the one-prime diagonal entry $\chi(2)2^{-1/2}$ is negative. Its 1×1 minor is already negative, so an unrestricted assertion of total nonnegativity of these transfer matrices is false. A positive state constructed as $TT^*/\operatorname{tr}(TT^*)$ discards phase information in this diagonal model: the nonzero character phases disappear from TT^* at real s, although the ramification zero mask remains. Thus positivity of that state cannot by itself recover character-sensitive Euler factors.

For $C_2 \times C_2$, the three order-two subgroups each admit two linear characters. All six induced characters have degree two; induction from the trivial subgroup gives degree four. Negative integer

coefficients do not fix the parity obstruction. Rational coefficients can change parity, which explains why rational and integral induction cannot be interchanged. For a monomial representation $\mathrm{Ind}_H^G \theta$, the appropriate Artin formalism yields a Hecke L-function over the fixed field of H; it does not automatically reduce to a product of Dirichlet functions over $\mathbb{Q}$. Every extension must retain the base field and infinity type.

C Local algebra and topology checks

For the upper-Iwahori order

$$\mathcal{O} = \begin{pmatrix} R & R \\ \pi R & R \end{pmatrix}, \quad S = \begin{pmatrix} 0 & \pi \\ 1 & 0 \end{pmatrix},$$

$E_{12} \in \mathcal{O}$, whereas $SE_{12}S^{-1} = \pi^{-1}E_{21} \notin \mathcal{O}$. The displayed shift orientation therefore does not normalize this order. Changing an orientation may repair this isolated calculation, but all subsequent products and gluing maps must be rechecked with that convention.

For $R = \mathbb{C}[[t]]$ and $\lambda \in \mathbb{C}^\times$, there is no unital $\mathbb{C}$-algebra homomorphism sending t to λI in a nonzero matrix algebra: $t - \lambda$ is a unit in R and would be sent to zero. Thus substituting a unit complex phase for a formal uniformizer is not a representation of this allowed complete DVR. An algebra of Laurent polynomials can admit unitary phase evaluation, but it is a different algebra and needs its own construction.

A dessin on a sphere with two vertices and three edges has three faces, by Euler's formula $V - E + F = 2$, not two. Repeating a vertex along a face boundary does not create a new primitive vertex idempotent. In particular, a three-vertex path algebra with arrows between distinct vertices cannot have the claimed surjection to a six-vertex radical-square-zero cycle algebra: compose with the target's semisimple quotient k^6. Arrows map to zero, since the orthogonal source idempotents force their endpoint supports to be disjoint in the commutative target. The image is generated by at most three orthogonal idempotents, so it cannot be all of k^6. A valid replacement may require a cover or unfolded category rather than a quotient of the stated algebra.

D Why weak convergence and boundary analogies are insufficient

Let P_n be projection onto the nth basis vector of ℓ^2. For every $x \in \ell^2$, $\|P_n x\| = |x_n| \to 0$, so $P_n \to 0$ strongly, but $\|P_n\|_1 = 1$ for every n. Moreover $\det(I - P_n) = 0$ while $\det(I) = 1$. Thus even strong operator convergence does not imply determinant convergence. Trace-norm control is a natural sufficient condition, and must be established for the actual operator family, locally uniformly in s, before invoking Fredholm determinant continuity.

For a Phragmén–Lindelöf argument, one needs uniform control of approximants as the approximation level changes. Polynomial growth of each individual polynomial has constants and degrees depending on N; it does not supply a common bound. Dividing by $(1 + |s|^2)^M$ produces a nonholomorphic expression, to which the maximum modulus principle does not apply. Dividing by $(1 + s^2)^M$ instead introduces poles at $s = \pm i$; a denominator must be holomorphic and nonvanishing on the actual domain. Even after selecting such a denominator, the necessary uniform boundary and horizontal-side estimates are substantive mathematical work.

Likewise, a bound on a density of geodesics is not a bound on a weighted analytic remainder. The full test-function dependence, prime powers, scattering, elliptic and parabolic contributions must be retained. On a cusped finite-area surface the infimum of the injectivity radius is zero. It cannot provide the positive global mesh bound claimed in the source completion argument.

E Spectral approximation can conceal the original hypothesis

Proposition E.1 (Canonical-product converse for the Riemann target)**.** *If RH holds, the entire order-one function ξ admits locally uniform approximations of the form in Theorem 2.1 with diagonal self-adjoint H_N.*

Proof. The Hadamard factorization of ξ, which has no zero at zero, is

$$\xi(s) = e^{a+bs} \prod_{\rho} (1 - s/\rho) e^{s/\rho},$$

with an enumeration for which the canonical product converges locally uniformly. Under RH every $\rho = 1/2 + i\gamma$ has real γ. Each finite product is an exponential of an affine function times a constant times $\prod_{j \le N}(s - \rho_j)$; absorb the nonzero constant into the exponential. Take $H_N = \mathrm{diag}(\gamma_1, \ldots, \gamma_N)$. □

Together with Theorem 2.1, this shows why existence of the approximants, without an independent arithmetic construction, may merely restate RH. It is not progress to import the unknown zero locations into the input spectrum and then prove that the constructed spectrum is real. This proposition is a classical consequence, not a novel proof of RH.

F Hermitian dilation and the spectral parameter

A common attempt to preserve arithmetic input while obtaining self-adjointness is the dilation, for square $T(s) \in M_d(\mathbb{C})$,

$$K(s) = \begin{pmatrix} 0 & T(s) \\ T(s)^* & 0 \end{pmatrix}.$$

It is Hermitian pointwise. Singular-value decomposition shows that its spectrum is $\{\pm\sigma_j(T(s)) : 1 \le j \le d\}$, with multiplicities. Consequently

$$\det(\lambda I - K(s)) = \prod_{j=1}^{d} (\lambda^2 - \sigma_j(T(s))^2), \qquad \det(-K(s)) = (-1)^d |\det T(s)|^2.$$

These formulas retain singular values and erase determinant phase. For a nonconstant holomorphic scalar f, $|f|^2$ cannot be holomorphic: a real-valued holomorphic function is constant, and constant modulus then forces f constant. Thus the zero-matching scalar $|f|^2$ is not a holomorphic replacement for f. Matrix nonholomorphy alone does not imply that every scalar expression formed from it is nonholomorphic; each proposed determinant identity must be examined separately. For example, $T(s) = \mathrm{diag}(e^s, e^{-s})$ is nonconstant but its dilation determinant is constantly one.

For the prime-diagonal model, the dilation eigenvalues are exactly $\pm|\chi(p)|p^{-\operatorname{Re} s}$. Both $\Im s$ and the nonzero character phases disappear. This cannot encode the proposed arithmetic zero ordinates. The source's parameter mismatch is visible where an s-dependent transfer is converted to a Hermitian/modular Hamiltonian and then used as though it had fixed eigenvalues in a characteristic polynomial [1], source lines 11465–11484 and 12200–12215; see also lines 13636–13643 for the explicit skew-Hermitian-part extraction. Its numerical code fixes a particular evaluation point instead, so the rigidity proposition does not itself refute that distinct operation. The earlier diagonal diagnostic obstruction addresses it directly.

A permitted architecture is a fixed self-adjoint H_N with spectral variable entering through $sI - (1/2 + iH_N)$. Alternatively one may use a holomorphic pencil that is Hermitian only on

its real parameter axis, but this condition does not force real spectral roots. Already the scalar polynomial $h(t) = t^2 + 1$ is real for real t, whereas $z - h(z) = 0$ has the nonreal solutions $(1 \pm i\sqrt{3})/2$. An infinite-dimensional operator model would additionally need explicit domains, closure, adjoint and determinant regularization. No finite-matrix rigidity statement here is asserted to settle variable-domain unbounded operator constructions.

G Positive pencils, resolvent control, and a Herglotz test

The positive generalized pencil is precisely a fixed self-adjoint construction in different coordinates. Put $H = B^{-1/2}AB^{-1/2} = H^*$. Then

$$A - zB = B^{1/2}(H - zI)B^{1/2},$$

and therefore

$$\det(A + i(s - 1/2)B) = i^d \det(B) \det\big(sI - (\tfrac{1}{2}I + iH)\big).$$

The positive metric B must be supplied by the arithmetic construction; its existence is not inferred from a desire for real roots.

There is also a quantitative test. If $B \geq bI$ with $b > 0$ and z is nonreal, then for every v,

$$\|(A - zB)v\| \, \|v\| \geq |v^*(A - zB)v| \geq |\Im z| v^*Bv \geq b|\Im z| \, \|v\|^2.$$

For $v \neq 0$, divide by $\|v\|$. In finite dimension this proves invertibility and the operator-norm bound

$$\|(A - zB)^{-1}\| \leq \frac{1}{b|\Im z|} = \frac{1}{b|\operatorname{Re} s - 1/2|}.$$

Uniform approximation arguments must track dimension, normalization and b_N; a bound for each individual pencil does not by itself provide uniform determinant convergence.

A nonlinear extension uses a matrix-valued Herglotz condition: if $M(z)$ is holomorphic upstairs and

$$\Im M(z) := \frac{M(z) - M(z)^*}{2i} > 0,$$

then $M(z)v = 0$ would contradict $v^*\Im M(z)v > 0$. Thus $\det M(z) \neq 0$ there. A reflected lower-half-plane extension $M(\bar{z}) = M(z)^*$ is likewise invertible. A quantitative lower bound $\Im M(z) \geq \eta(z)I > 0$ gives $\|M(z)^{-1}\| \leq 1/\eta(z)$ by the same argument. For the positive linear pencil the Herglotz sign is $M(z) = zB - A$, not $A - zB$. Semidefinite positivity alone allows $M(z) = \operatorname{diag}(z, 0)$, which is always singular. Without reflection, $M(z) = z + i$ is strictly Herglotz upstairs but has a lower-half-plane zero. Possible real-axis poles and the arithmetic determinant identity remain separate obligations.

For a candidate normalized identity written in the z-coordinate, $F_N(z)/g_N(z) = \det(A_N - zB_N)$ with a prescribed nowhere-zero g_N, a necessary diagnostic, for $d_N > 0$, is

$$-\partial_z \log(F_N/g_N) = \operatorname{tr}\big((A_N - zB_N)^{-1}B_N\big) = \sum_{j=1}^{d_N} \frac{1}{\lambda_{j,N} - z}.$$

Its imaginary part is strictly positive for $\Im z > 0$. A certified violation rejects that particular determinant identity; finitely many successful samples cannot prove it. One must not choose g_N using unknown zero locations or assume this Herglotz positivity as a substitute for establishing the arithmetic bridge.

H Quantitative arithmetic approximation and prime regularization

For the nonprincipal-character approximants, a compact set $K \subset \{\operatorname{Re} s > 0\}$ has $\delta = \min_K \operatorname{Re} s > 0$, $M = \max_K |s|$, and $G = \max_K |(q/\pi)^{(s+a)/2}\Gamma((s+a)/2)| < \infty$. The completed approximation error is at most $GB(1 + M/\delta)N^{-\delta}$ uniformly on K. This is a fixed-character compact bound, not a uniform-in-conductor GRH estimate. To verify the Abel formula's sign, first sum between integers $N < T$:

$$\sum_{N<n\le T} \chi(n)n^{-s} = A(T)T^{-s} - A(N)N^{-s} + s\int_N^T A(x)x^{-s-1}\,dx.$$

Then $A(T)T^{-s} \to 0$. Bounding $|A|$ by B gives exactly the main-text estimate. For zeta, $\lfloor x \rfloor = x - \{x\}$ in the same summation formula yields

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\}x^{-s-1}\,dx.$$

The finite sum formula gives the stated remainder after subtracting Z_N. The integral and its derivatives converge uniformly on compact subsets of the positive half-plane, since logarithmic powers are integrable against $x^{-1-\delta}$. This justifies holomorphy and continuation. Multiplying the remainder by $s(s-1)\pi^{-s/2}\Gamma(s/2)/2$ gives the compact-uniform ξ_N error, including at $s = 1$. In contrast, multiplying a finite Euler product by $s(s-1)$ would force a zero at 1 for every truncation, although $\xi(1) = 1/2$.

A second exact interface isolates what a prime-based repair would need. For a nonprincipal primitive χ, define on $\operatorname{Re} s > 1$

$$P_\chi(s) = \sum_p \chi(p)p^{-s}, \qquad R_\chi(s) = \sum_p \sum_{k\ge 2} \frac{\chi(p)^k}{kp^{ks}}.$$

On each compact subset of $\operatorname{Re} s > 1/2$, the second series converges absolutely and uniformly. Indeed, for $\operatorname{Re} s \ge \delta > 1/2$,

$$\sum_{k\ge 2} \frac{|\chi(p)|^k}{kp^{k\delta}} \le \frac{p^{-2\delta}}{1 - 2^{-\delta}},$$

and the prime sum is dominated by the convergent integer series $\sum_{n\ge 2} n^{-2\delta}$. Thus R_χ is holomorphic there. On $\operatorname{Re} s > 1$, $L(s,\chi) = \exp(P_\chi(s) + R_\chi(s))$.

Proposition H.1 (The missing prime-linear term)**.** *For a nonprincipal primitive character, P_χ has a single-valued holomorphic continuation to $\operatorname{Re} s > 1/2$ if and only if $L(s,\chi)$ is zero-free there.*

Proof. If such a continuation exists, its exponential with R_χ agrees with L on $\operatorname{Re} s > 1$ and hence on the connected half-plane by the identity theorem. The exponential is nonzero. Conversely, if L is nonzero, the simply connected half-plane admits a holomorphic logarithm of L. Choose its additive constant to agree with the absolutely convergent Euler logarithm on $\operatorname{Re} s > 1$. Subtract R_χ to obtain the required continuation of P_χ. □

This proposition isolates a precise analytic obligation; it does not prove it. Regularizing away the terms $k \ge 2$ supplies no proof that the missing prime-linear function extends. The principal zeta case must first remove the pole, for example by using $(s-1)\zeta(s)$; an unqualified logarithm of ζ cannot be holomorphic through 1. The functional equation relates a fixed character's zeros across the critical line after complex conjugation, so a full right-half-plane nonvanishing theorem would give the desired Dirichlet GRH conclusion.

I Exact arithmetic behind the finite-level zero certificate

This section gives all parameters needed to reproduce Theorem 7.1. Write $C = |\operatorname{Re} c| + |\Im c| < 12$. Use the following rational logarithm approximations ℓ_n and error bounds δ_n:

n	ℓ_n	δ_n
2	$6931471805599453/10^{16}$	10^{-15}
4	$13862943611198906/10^{16}$	$2 \cdot 10^{-15}$
5	$16094379124341004/10^{16}$	10^{-15}

For $y = (n-1)/(n+1)$, set

$$L_n = 2\sum_{k=0}^{59} \frac{y^{2k+1}}{2k+1}, \qquad T_n = \frac{2y^{121}}{121(1-y^2)}.$$

The positive series for $\log n$ gives $L_n \le \log n \le L_n + T_n$. Exact rational comparisons give $\ell_n - \delta_n \le L_n$ and $L_n + T_n \le \ell_n + \delta_n$. They also give $0 < \log 2 < 1$ and $0 < \log 4, \log 5 < 2$.

Define Gaussian rationals

$$E_n = \sum_{k=0}^{120} \frac{(-c\ell_n)^k}{k!}, \qquad \tau = \frac{3^{24}24^{121}}{121!} < 10^{-18}, \qquad \eta_n = \tau + 2C\delta_n.$$

Since $|c\ell_n| \le C\ell_n < 24$, the exponential series tail is at most $e^{24}24^{121}/121! < \tau$. Here $e < 3$ follows directly from its series. Also

$$|e^{-c\log n} - e^{-c\ell_n}| \le e^{-\operatorname{Re}(c)\ell_n}|c|\delta_n e^{|c|\delta_n} \le 2C\delta_n,$$

because $\ell_n, \operatorname{Re} c > 0$, $C\delta_n \le 1/2$, and $e^u \le (1-u)^{-1} \le 2$ for $0 \le u \le 1/2$. Thus $|n^{-c} - E_n| \le \eta_n$.

Put

$$F = 1 - E_2 + E_4 - E_5, \qquad D = \ell_2 E_2 - \ell_4 E_4 + \ell_5 E_5.$$

All entries are exact Gaussian rationals. With $\|(a,b)\|_1 = |a| + |b|$, the script checks the rational inequalities

$$\|F\|_1 + \eta_2 + \eta_4 + \eta_5 < 10^{-6}, \qquad |\operatorname{Re} D| - \sum_{n\in\{2,4,5\}} (\delta_n + \ell_n\eta_n) > 1.$$

The first bounds $|f(c)|$. For the second, note that $|n^{-c}| \le 1$ and

$$|(\log n)n^{-c} - \ell_n E_n| \le \delta_n + \ell_n\eta_n.$$

This proves $|f'(c)| > 1$. On the entire closed disk, $\operatorname{Re} s > 0$, so

$$|f''(s)| \le (\log 2)^2 + (\log 4)^2 + (\log 5)^2 < 1 + 4 + 4 = 9.$$

Integrating the second derivative along the line segment from c to s justifies the Taylor remainder bound $9|s-c|^2/2$. The exact strict boundary margin is

$$r - 10^{-6} - \tfrac{9}{2}r^2 = \frac{19791}{200000000} > 0.$$

Consequently $|f(s)|$ is greater than that margin on $|s-c| = r$. Any zero-free holomorphic positive-pencil model G on a neighborhood of this disk must have boundary error at least $\min_{|s-c|=r}|f(s)|$; otherwise another application of Rouché would contradict its zero count. This is a quantitative

rejection of a prescribed finite-level model, not a proof about the zero locations of the limiting L-function.

The certificate program `scripts/check_arithmetic_zero.py` uses only Python integer and Fraction arithmetic. A separate independent exact-arithmetic reproduction passed. Lean verifies the disk's location and the last rational margin; the logarithm/exponential error estimates and the Rouché inference remain explicit prose proofs, not claims of a complete kernel-checked zero-existence theorem.

J Research tasks with mathematical acceptance criteria

Track A: repair the determinant interface. Start with the exact diagonal Euler transfer on $\operatorname{Re} s > 1$. Determine which regularized operator determinant could continue the reciprocal Euler product without poles or zeros being silently exchanged. Prove a relation to a self-adjoint operator before replacing the transfer by its Hermitian part. Acceptance requires an identity of functions, not similarity of zero plots. The displayed 2×2 example rejects unrestricted determinant preservation immediately.

Track B: trace norms and normal families. Choose a connected domain containing an Euler-product overlap and the critical strip, after removing the known poles in the target. Supply estimates uniform in N on every compact set. Either construct a trace-class holomorphic family with uniform trace-norm convergence or prove the hypotheses of Proposition 2.3. The divergence in Theorem 3.3 prevents an absolute prime-sum shortcut. Any cancellation estimate must be independently proved without a zero-free region equivalent to the desired conclusion.

Track C: positivity and canonical products. Compare the exact spectral quadratic form with Weil positivity and the Li coefficients for the chosen L-function family. First state the test-function space and regularization, then prove equality of the forms. Positivity of a finite matrix or of finitely many Li coefficients is insufficient; a uniform infinite-family statement is required. Canonical products provide a convergence benchmark, but the preceding converse explains the circularity risk of using zeros as input.

Track D: make formalization match mathematics. The standalone Lean companion checks exact integer and finite-matrix obstructions. A separate mathlib layer now verifies complex Euler-factor nonvanishing, a local boundary-convergence theorem, and holomorphic Hermitian rigidity. It does not yet formalize the Dirichlet partial-sum convergence or an arithmetic self-adjoint operator. Further formalization must use the actual standard L-functions and explicit analytic domains. An abstract theorem of the form "if an assumed bridge implies GRH, then GRH" is logically valid but would not verify the bridge. Print every theorem's axioms and review its full type and dependencies, not merely its name.

K Computational examples and reproducibility

The Python checks use integers and `fractions.Fraction`, with no random choices, model weights, scientific datasets or floating-point zero finding. They reproduce the Iwahori witness at $\pi = 2$, the determinant change $2 \to 1$, the parity obstruction, and the vanishing diagonal model. The Lean statements verify selected finite calculations and the parity argument in the kernel of a pinned Lean 4 toolchain. Neither executable claims to verify the full complex-analytic arguments in this article.

Both editions share the same core source. The longer edition adds these calculations and research interfaces; extra exposition does not strengthen an unproved hypothesis into a theorem. The source manifest and audit coverage record distinguish full text review, focused review, and unverified dependencies. Publication readiness also requires independent mathematical review, confirmed authorship, and journal-specific preparation.

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